Simulation Transcript
Secure an IBM Cloudant database

Darren, the senior security analyst, welcomes you to the project. Here, you have IBM Cloud open and ready to use. Select the **Next arrow** to continue.

1. Darren explains that the first step is to create an IBM Cloudant database on IBM Cloud. IBM Cloudant databases are ideal for web, mobile, and serverless cloud applications and the development team requested this type of database from your team. Select the **Next arrow** to continue.

2. Darren asks you to change the name in the **Instance name** field to "usr-cldnt-database" and leave all other values as their defaults. Select the **Create** button to create the database.

3. Darren thanks you for setting up the database and explains that the next step is to prepare for the security check. Because the database will be storing user information, your team needs to ensure that it's using encryption and a secure authentication method. First, you will perform a visual check of the database settings. Select the **Next arrow** to continue.

4. Darren asks you to open the resources page of the IBM Cloudant database you just created to verify two settings. Under the **Name** column, select **usr-cldnt-database** from the list of databases.

5. Darren asks you to verify two settings to ensure the database is secure. Notice that the **Authentication methods** is using IBM Cloud IAM. This ensures that only authorized users can access the database. Also notice that **Disk encryption** is set to Yes, which ensures that the data in the database is securely stored and transported. Select the **Next arrow** to continue.

6. Your next task is to create a cloud object storage instance. Darren explains that to store the results from the compliance and security tests you'll be running, you need to set up this object storage service. This type of storage is best for files and other unstructured data. Select the **Next arrow** to continue.

7. You find the Cloud Object Storage service in the IBM catalog. Take a moment to review all the default settings for the instance you're creating. Notice that you're creating this storage on the IBM Cloud, but it's possible to use object storage from an on-premises source or even from another cloud provider. Select the **Create** button.

8. Darren explains that you've created the cloud object. Now you need to create a bucket to store any files that will be put in object storage. For this project, the result of your security test will be stored here. Select the **Create bucket** button.

9. Darren says you can use a predefined bucket for this project. He explains that using this option will optimize the bucket type for you, so you don't have to configure a lot of options. Under the **Predefined buckets** section, select the arrow in the **Quickly get started** block.

10. Darren tells you that you can leave the default bucket name as it is. Select the **Next** button.

11. Because you're not uploading files at this point, you can just create your bucket. Select the **View bucket configuration** button.

12. Darren thanks you for creating the cloud object storage and tells you that you're now ready to create the Cloudant tests and store them in the cloud object storage you just created. Select the **Next arrow** to continue.

13. Before you can run any security tests, Darren attaches the cloud object storage you created earlier. He attaches the storage by configuring the settings in the Security and Compliance tool. Select the **Next arrow** to continue.

14. Darren explains that you're ready to create some tests that you can run against the IBM Cloudant database to make sure it's ready for the development team. He points out on the Security and Compliance page that the service enables you to create a control that will run specific tests against services on IBM Cloud. He asks you to start by creating a control library that will store the controls. A **control** is similar to a policy in that it defines the security measure your organization requires. Darren has selected the **Controls** menu item. Under **Controls**, select **Control libraries**.

15. Darren asks you to create a new control library that will store the IBM Cloudant controls. You open the **Control libraries** menu item so you can create a new control library. Select the **Create** button to create the control.

16. Darren asks you to name the control library. Place your cursor in the **Name** field and type the following text. Then, press **Enter**.

Cloudant Database Controls

17. Darren asks you to give the control library a meaningful description. Place your cursor in the **Description** field and type the following text. Then, select the **Create** button.

A set of controls to ensure our Cloudant database is secure

18. Darren thanks you for creating the control library. He explains that the control library needs **assessments**—the actual tests the tool will perform against the database. He creates a new control called **Cloudant developer control**. Darren finishes the steps to add assessments to it. Select the **Next arrow** to continue.

19. Darren asks you to examine each of the Cloudant control assessments. Notice that two of them are related to secure communication. The last one in the list is related to a specific type of encryption that corporations use when they want to tightly control particular areas of their security. Select the **Next arrow** to continue.

20. Darren explains that you'll run the IBM Cloudant tests that will check that the database is enabled with encryption and is accessible only through HTTPS, which is a secure way to send data over the internet. Select the checkboxes for **Check whether Cloudant is enabled with encryption** and **Check whether Cloudant is accessible only through TLS 1.2 or higher** and select **Create**.

21. Now that you've added the assessments, you can create the control. On the **Create a control** page, select **Create**.

22. Darren thanks you for your help and tells you that you're almost ready to run the assessments against the IBM Cloudant database. In order to run the tests, the Security and Compliance service needs a profile under which to run. Darren completes this last step. Select the **Next arrow** to continue.

23. Darren tells you that now that the profile has been set up, it's time to create an attachment. An attachment is joining a profile to a set of assessments. Once you join a profile with assessments, the IBM Cloudant tests will run and you can view the results. Select **Create**.

24. Here's the **Profile and scope** page. Darren tells you that you need to select a scope that includes the IBM Cloudant database you created earlier. You'll use the default scope for this project. Under **Scope**, select **Default** from the drop down menu, and then select the **Next** button.

25. This is the **Parameters** page. There aren't any parameters you need to set or edit for this assessment. You can skip this and select the **Next** button.

26. Here's the **Scan settings** page. Darren tells you that you can leave the scan settings to their defaults. The scan setting defaults will run a scan on your IBM Cloudant database once every day and show you a report. This will help you know if something critical has changed on your database that might compromise the data in it. You'll be able to see the results in the Security and Compliance dashboard. Select the **Next** button.

27. This is the **Review** page. Darren reviews the attachment settings and thanks you for your work. He instructs you to create the attachment that will start the scan. After the scan runs, the tool will create a file that stores the result of the tests and place it in the object storage bucket you created in a previous step. Select the **Create** button.

28. Darren congratulates you on a successful scan. He asks you to review the results. Notice the green check marks next to each of the assessments. This means your IBM Cloudant database passed the tests. Nice work! Select the Next arrow to continue.

You successfully created a secure IBM Cloudant database. You reviewed the settings on the database that showed that the database is secure. You then created controls that test the database to assess that the database uses the HTTPS protocol and that it is encrypted. You attached a profile to these controls that regularly tests the database to ensure nothing has changed.